

CSS Notes + 30 Practice Problems + Interview Questions

This PDF is designed for 2+ years frontend developers preparing for interviews. It contains CSS notes from basic to advanced level, 30 practice problems, and most asked interview questions.

1. What is CSS?

CSS (Cascading Style Sheets) is used to style HTML elements and control layout, colors, spacing, animations, responsiveness, and visual appearance.

2. CSS Syntax

```
selector {  
  property: value;  
}
```

3. Types of CSS

Inline CSS, Internal CSS, and External CSS.

4. Selectors

Element selector, class selector, id selector, universal selector, attribute selector, pseudo selectors.

5. Box Model

Every HTML element consists of content, padding, border, and margin.

6. Positioning

static, relative, absolute, fixed, sticky.

7. Display Properties

block, inline, inline-block, flex, grid, none.

8. Flexbox

Used for one-dimensional layouts. Important properties: justify-content, align-items, flex-direction, gap.

9. CSS Grid

Used for two-dimensional layouts. Important properties: grid-template-columns, rows, gap, place-items.

10. Responsive Design

Use media queries, flexible layouts, percentages, rem/em units.

11. Units

px, %, em, rem, vh, vw.

12. Specificity

Inline > ID > Class > Element selector.

13. Pseudo Classes

:hover, :focus, :nth-child(), :first-child.

14. Pseudo Elements

::before, ::after, ::placeholder.

15. Transitions

Smooth property changes using transition property.

16. Animations

Use @keyframes and animation property.

17. Z-index

Controls stacking order of positioned elements.

18. Overflow

visible, hidden, scroll, auto.

19. Object-fit

Controls how image/video fits inside container.

20. CSS Variables

```
:root {  
  --primary-color: blue;  
}
```

21. Media Queries

```
@media (max-width: 768px) {  
  body {  
    background: lightgray;  
  }  
}
```

22. Transform

translate, scale, rotate, skew.

23. CSS Best Practices

Use reusable classes, maintain consistent naming, avoid unnecessary nesting.

24. BEM Naming

Block__Element--Modifier naming convention.

25. Accessibility

Maintain proper contrast, focus states, readable font sizes.

26. Tailwind CSS

Utility-first CSS framework widely used in React and frontend projects.

27. Performance Optimization

Minimize reflows, optimize animations, reduce unused CSS.

28. Common Layout Problems

Centering divs, equal height cards, sticky footer, responsive navbar.

29. CSS Interview Focus

Flexbox, Grid, positioning, specificity, responsiveness, animations.

30. Real World Frontend Usage

CSS is heavily used in dashboards, ecommerce websites, admin panels, landing pages, and mobile responsive apps.

30 CSS Practice Problems

- 1 Center a div horizontally and vertically.
- 2 Create a responsive navbar.
- 3 Build a product card using Flexbox.
- 4 Create a pricing section using CSS Grid.
- 5 Build a responsive image gallery.
- 6 Create hover effects on buttons.
- 7 Build a loading spinner using CSS animation.
- 8 Create a dropdown menu.
- 9 Make a sticky navbar.
- 10 Build a responsive dashboard layout.
- 11 Create a card flip animation.
- 12 Build a modal popup.
- 13 Create a tooltip using pseudo-elements.
- 14 Build a responsive login page.
- 15 Create a CSS-only accordion.
- 16 Design a timeline component.
- 17 Build a responsive footer.
- 18 Create skeleton loading UI.
- 19 Build a notification toast component.
- 20 Create a progress bar animation.
- 21 Design a chessboard using Grid.
- 22 Create an animated sidebar.
- 23 Build a responsive table.
- 24 Create equal-height cards.
- 25 Design a dark mode toggle UI.
- 26 Build a masonry layout.
- 27 Create a typing text animation.
- 28 Build a responsive portfolio section.
- 29 Create a glassmorphism card.
- 30 Build a complex admin dashboard layout.

Most Asked CSS Interview Questions (2+ Years Experience)

- Difference between Flexbox and Grid
- Explain CSS specificity

- Difference between relative and absolute positioning
- What is z-index?
- Difference between em and rem
- How does box-sizing work?
- Difference between visibility hidden and display none
- How to center a div?
- What are pseudo classes and pseudo elements?
- What is BEM naming?
- Difference between inline, block, and inline-block
- How media queries work?
- What causes reflow and repaint?
- How to optimize CSS performance?
- Difference between transition and animation
- What is object-fit?
- How sticky positioning works?
- What are CSS variables?
- Explain stacking context
- What are accessibility best practices in CSS?